

# Automated Web-based Security Framework (AWSF)

## 1. Project Abstract and Goal

The Automated Web-based Security Framework (AWSF) is a Final Year Project focused on **streamlining cybersecurity reconnaissance and vulnerability assessment**. The core goal is to eliminate the inefficiency and complexity associated with manually executing multiple command-line security tools (such as **Nuclei, Findomain, and Amass**) by providing a single, intuitive **User Interface (UI)**. The framework translates user selections into automated, executable shell commands, processes the output, and presents results in a structured, accessible format.

## 2. Problem Statement

Current security workflows require analysts to manually issue multiple, disparate terminal commands for a single target, leading to significant time wastage and high barriers to entry. Furthermore, outputs from various tools are inconsistent and unstructured, complicating analysis and result management.

## 3. Technical Architecture

The AWSF follows a robust, three-tier architecture leveraging modern web and Python technologies:

- **Frontend (UI):** Built using **HTML, CSS, and JavaScript**. Its primary function is to capture user input (target domain and selected tools) and display final results in a clean, consistent format.
- **Backend (API & Orchestration):** Powered by **FastAPI (Python)**. This layer manages all application logic, validates user requests, orchestrates tool execution, and handles data parsing and persistence.
- **Tool Execution Layer:** External security tools are installed on the backend host and executed via Python's built-in **subprocess** module, which allows the FastAPI API to dynamically run command-line tools.

## 4. Key Functionalities

The framework includes essential features to ensure a seamless and efficient workflow:

| Feature           | Name Category | Description                                        |
|-------------------|---------------|----------------------------------------------------|
| Unified Interface | UI/UX         | A single, intuitive web dashboard for managing all |

|                            |                 |                                                                                                                                            |
|----------------------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------|
|                            |                 | security assessments and tools.                                                                                                            |
| Target & Tool Selection    | Input/Control   | Allows the user to input a single target domain and select multiple tools for execution at once.                                           |
| Asynchronous Execution     | Performance     | Scans run as <b>non-blocking background tasks</b> on the server, ensuring the UI remains responsive for the user.                          |
| Orchestration & Chaining   | Backend Logic   | The backend intelligently manages the sequential or parallel execution of selected tools for complex workflows.                            |
| Dynamic Command Generation | Backend Logic   | The frontend periodically checks the API for the scan's status using a <b>Job ID</b> until completion.                                     |
| Real-time Status Polling   | Communication   | The frontend periodically checks the API for the scan's status using a <b>Job ID</b> until completion.                                     |
| Structured Output          | Data Processing | Parses raw, inconsistent tool output and displays results in a clean, centralized, and consistent format on the web page.                  |
| Downloadable Report        | Reporting       | Allows users to generate and download a comprehensive, standardized PDF or HTML report summarizing all findings for professional delivery. |
| Persistent Results         | Data Storage    | All scan outputs are saved on the server for review and historical record-keeping.                                                         |